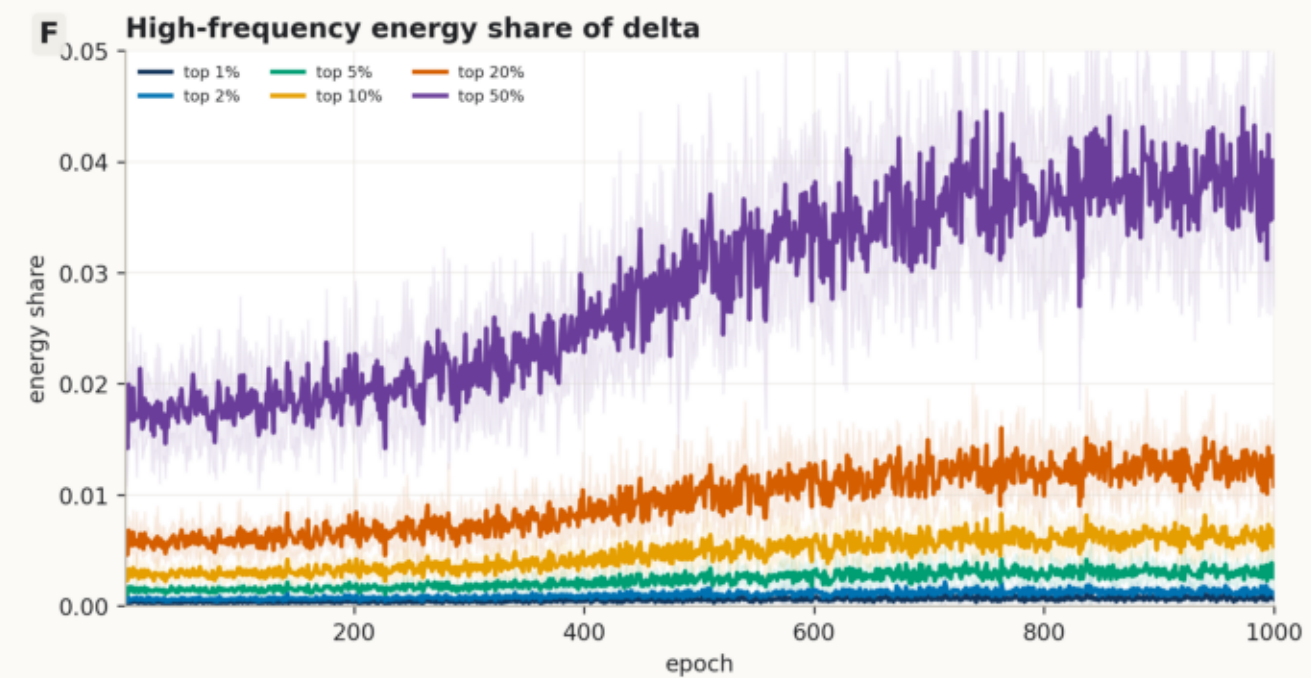
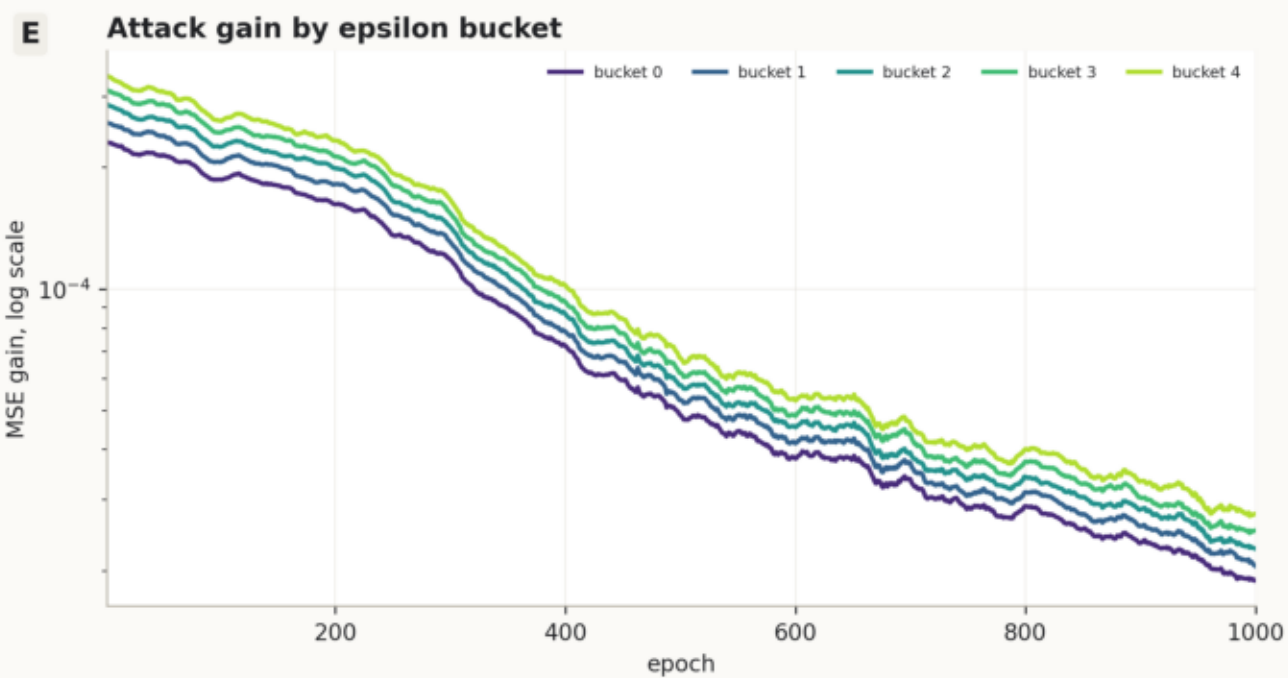
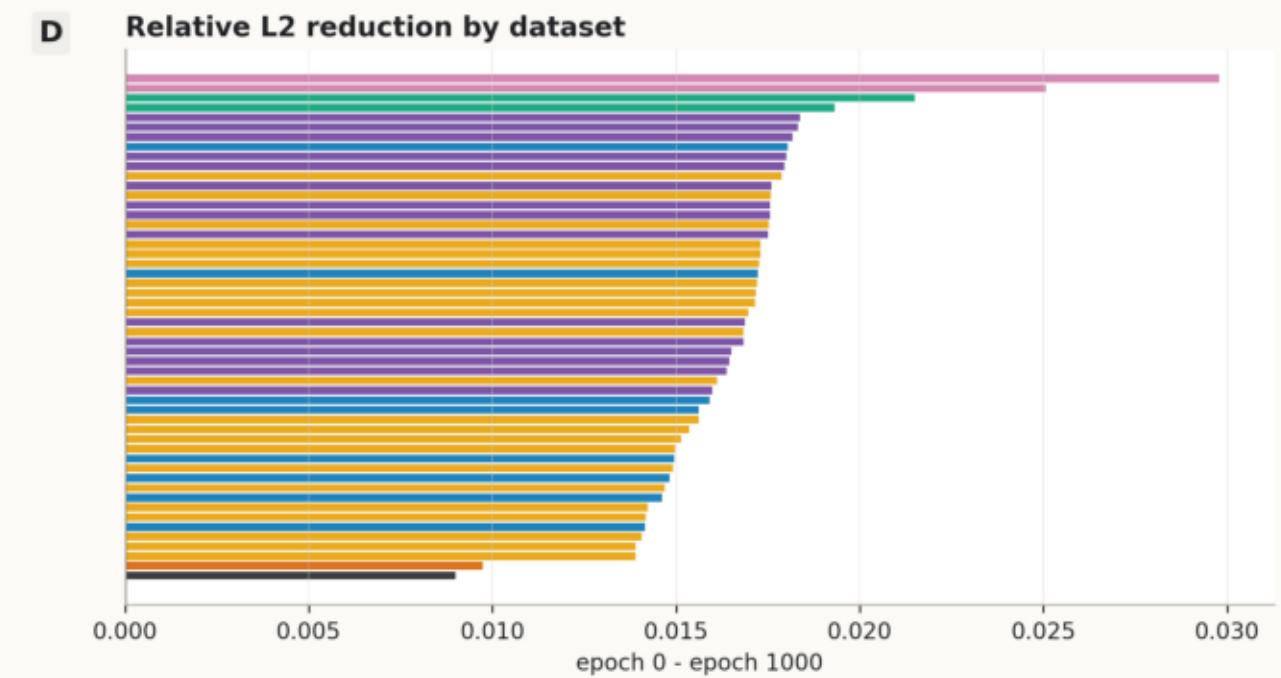
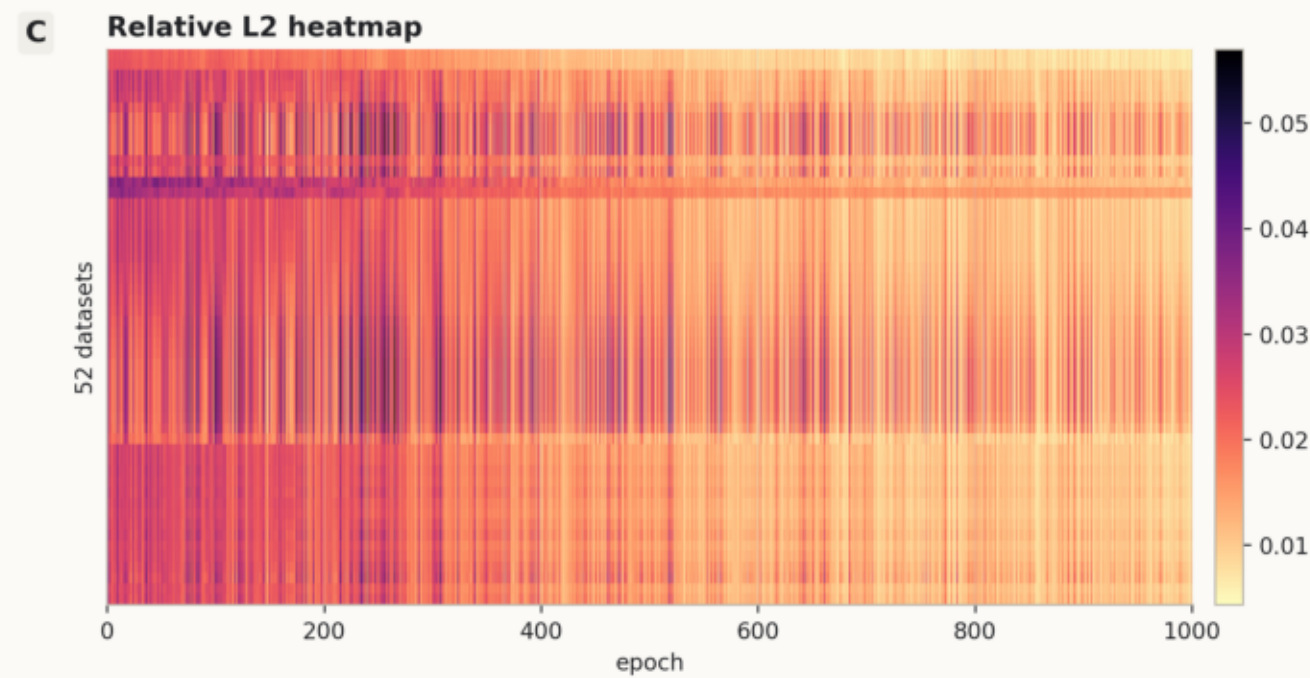
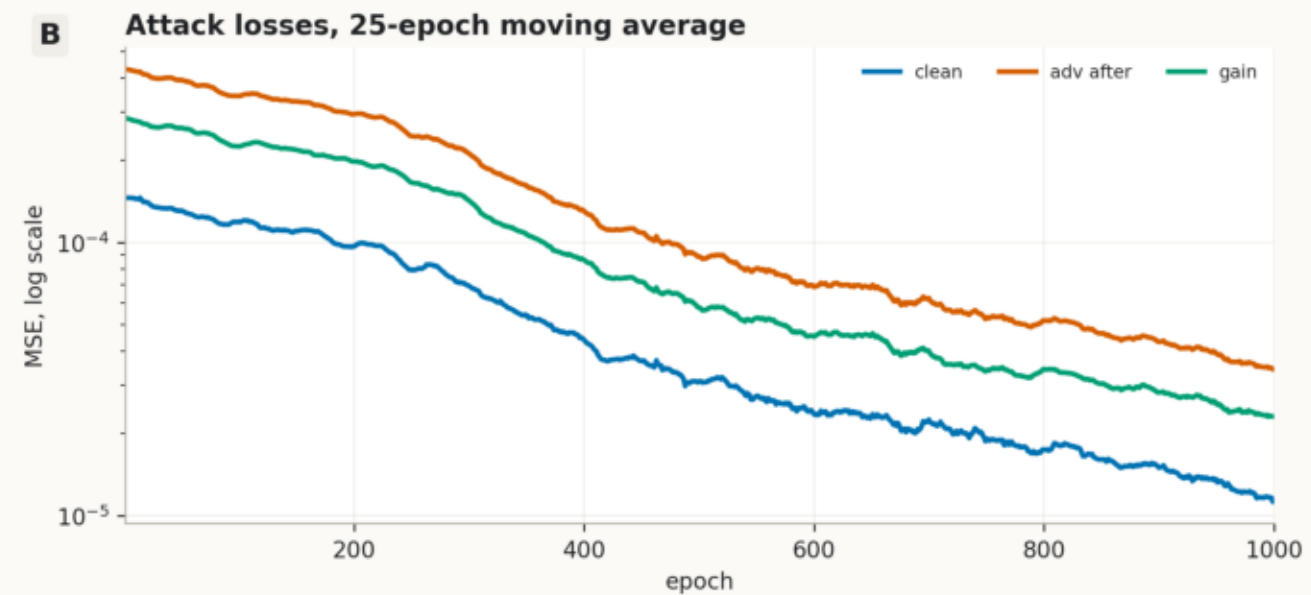
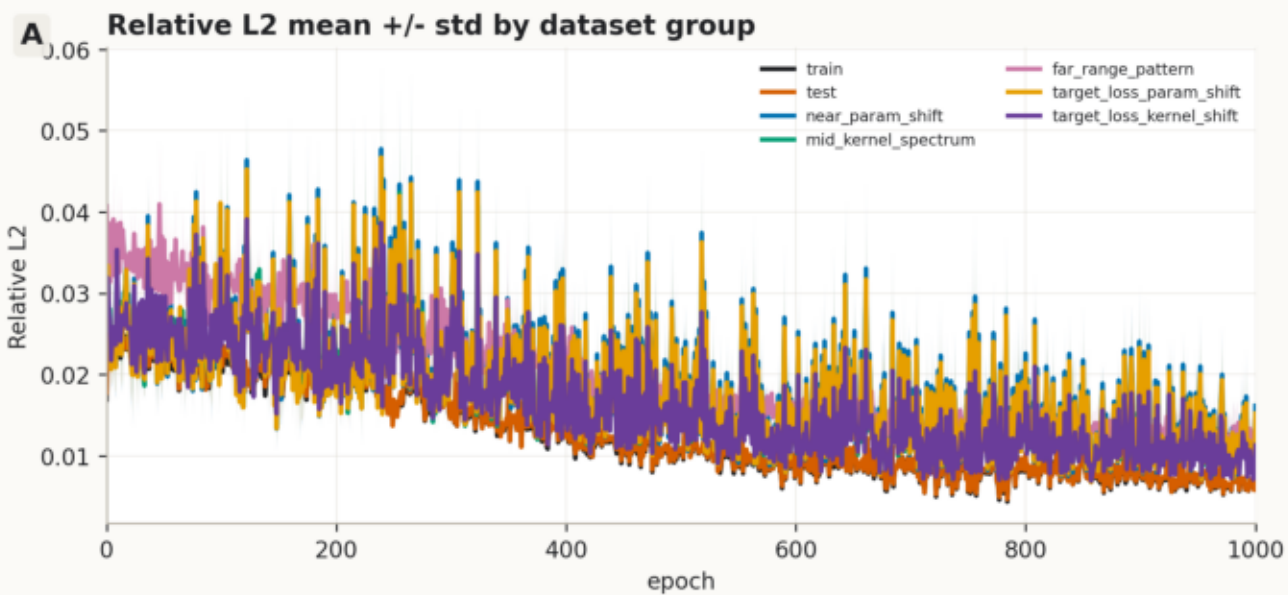


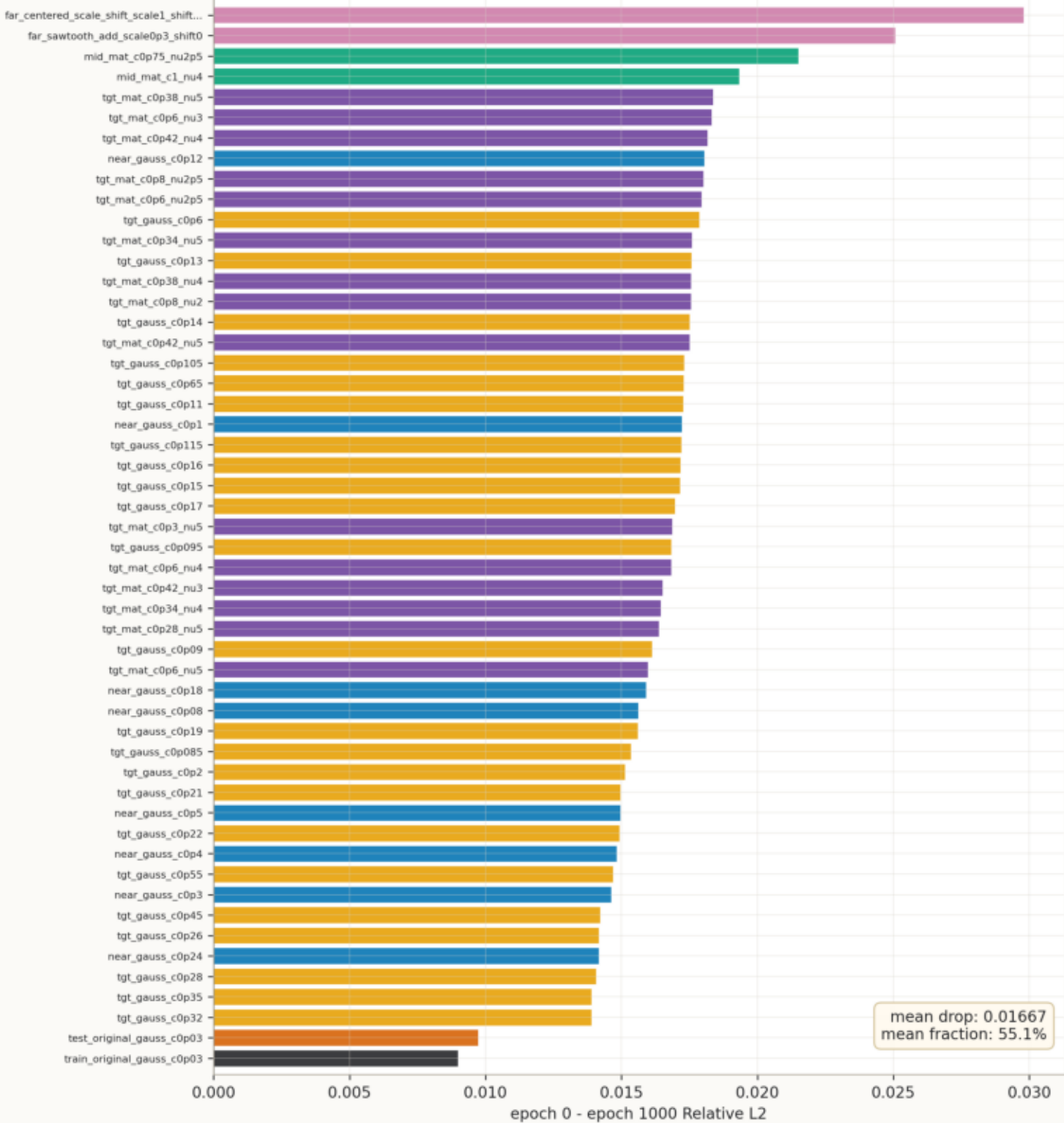
Burgers zero adversarial training: polished diagnostic dashboard

52/52 datasets improved | mean Relative L2 0.0305 -> 0.0138 | mean fractional drop 55.1%

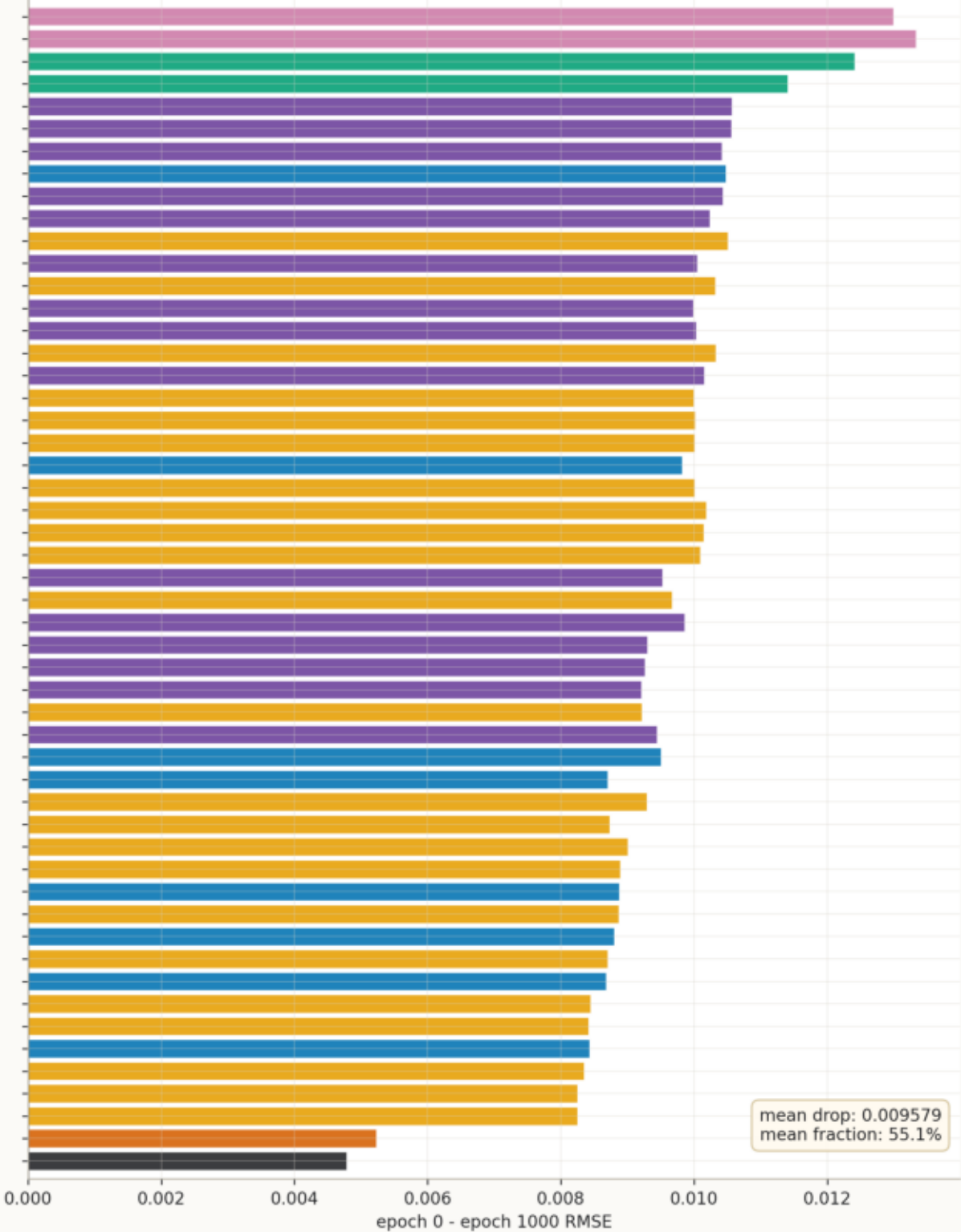


Burgers ADV-only training: loss reduction by dataset

Relative L2: 52 / 52 datasets improved

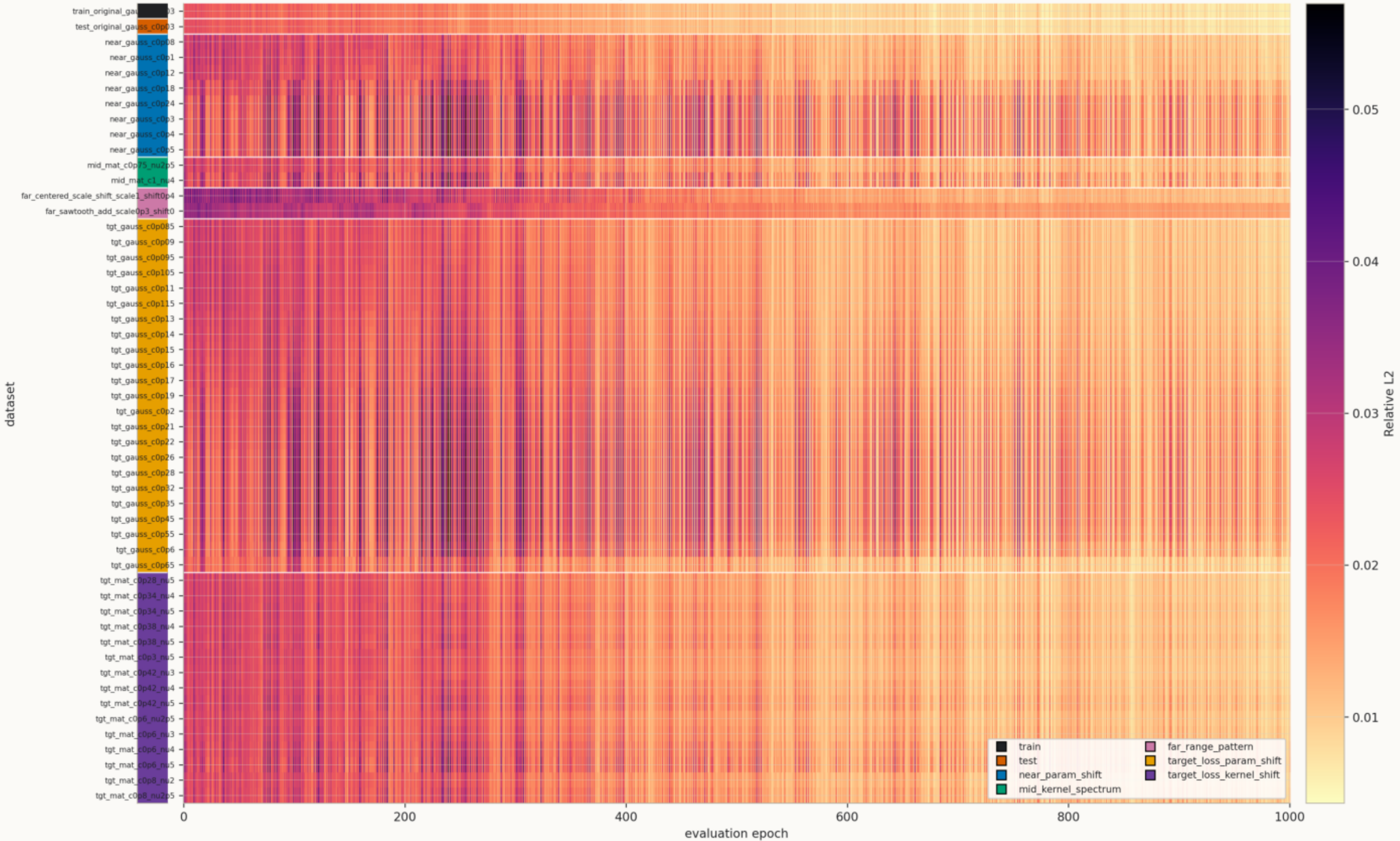


RMSE: 52 / 52 datasets improved



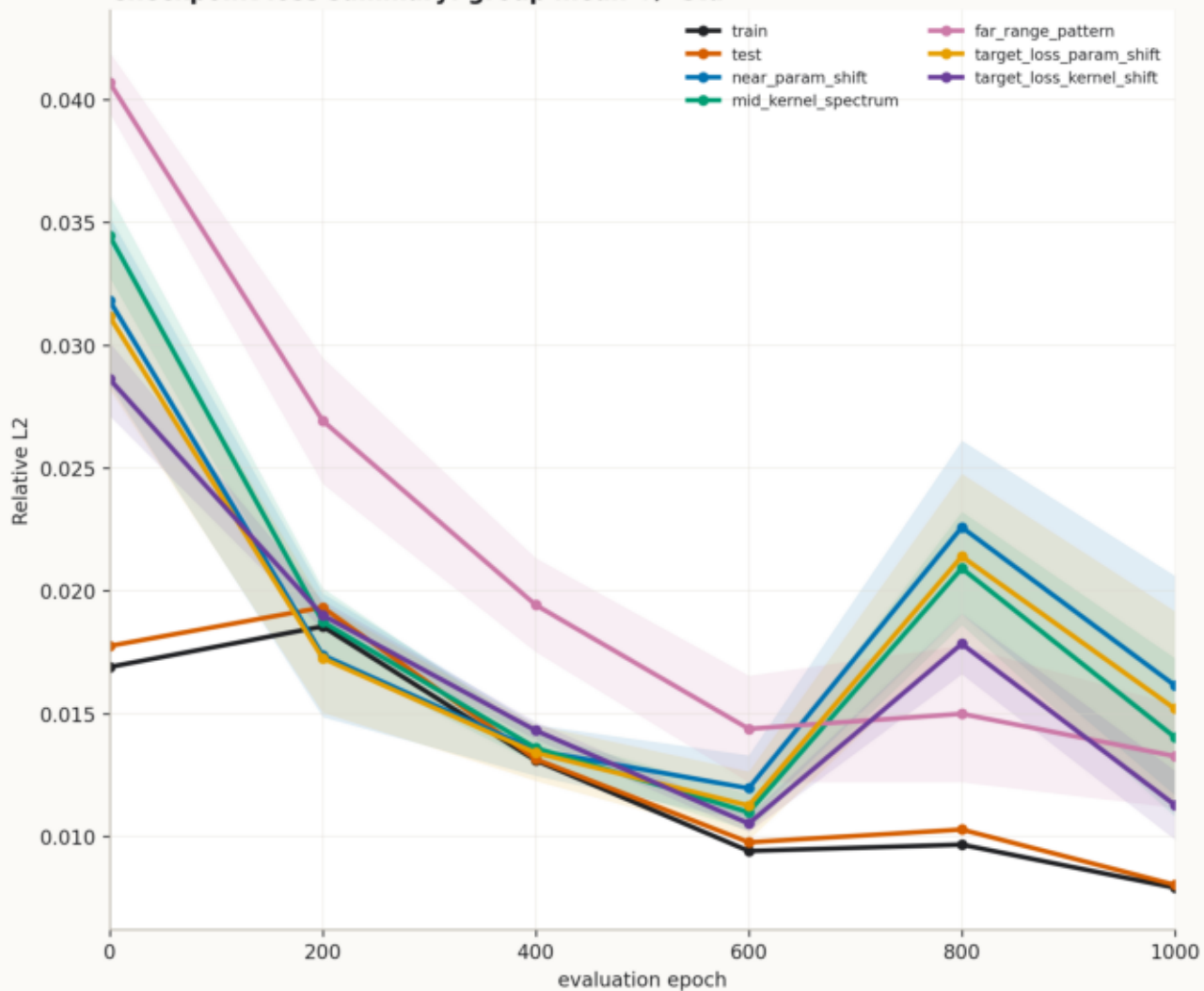
train test near_param_shift mid_kernel_spectrum far_range_pattern target_loss_kernel_shift target_loss_param_shift

Relative L2 across 52 datasets and 1001 evaluation epochs

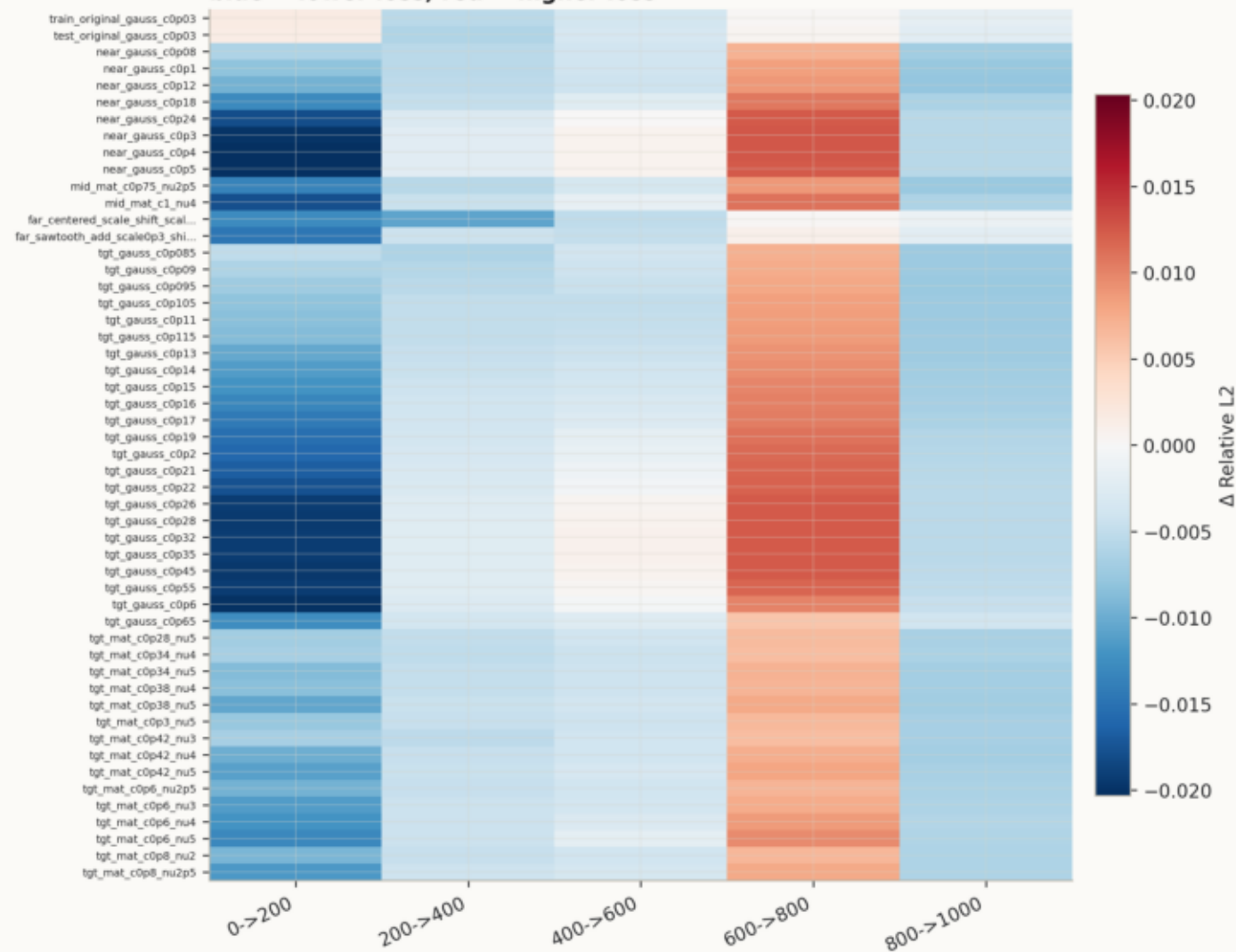


Burgers ADV-only: checkpoint loss behavior

Checkpoint loss summary: group mean $\pm$ std

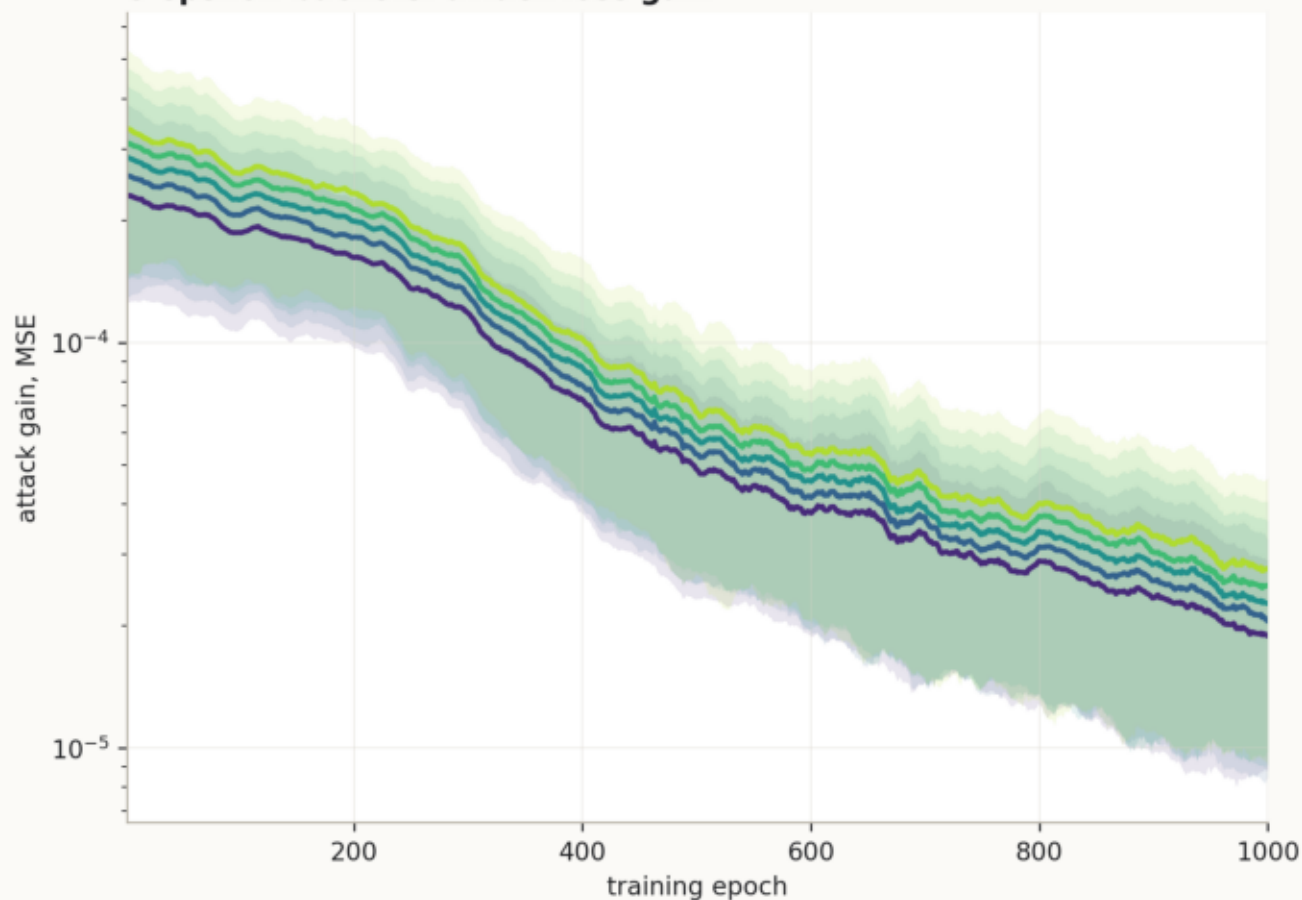


Checkpoint-to-checkpoint change
blue = lower loss, red = higher loss

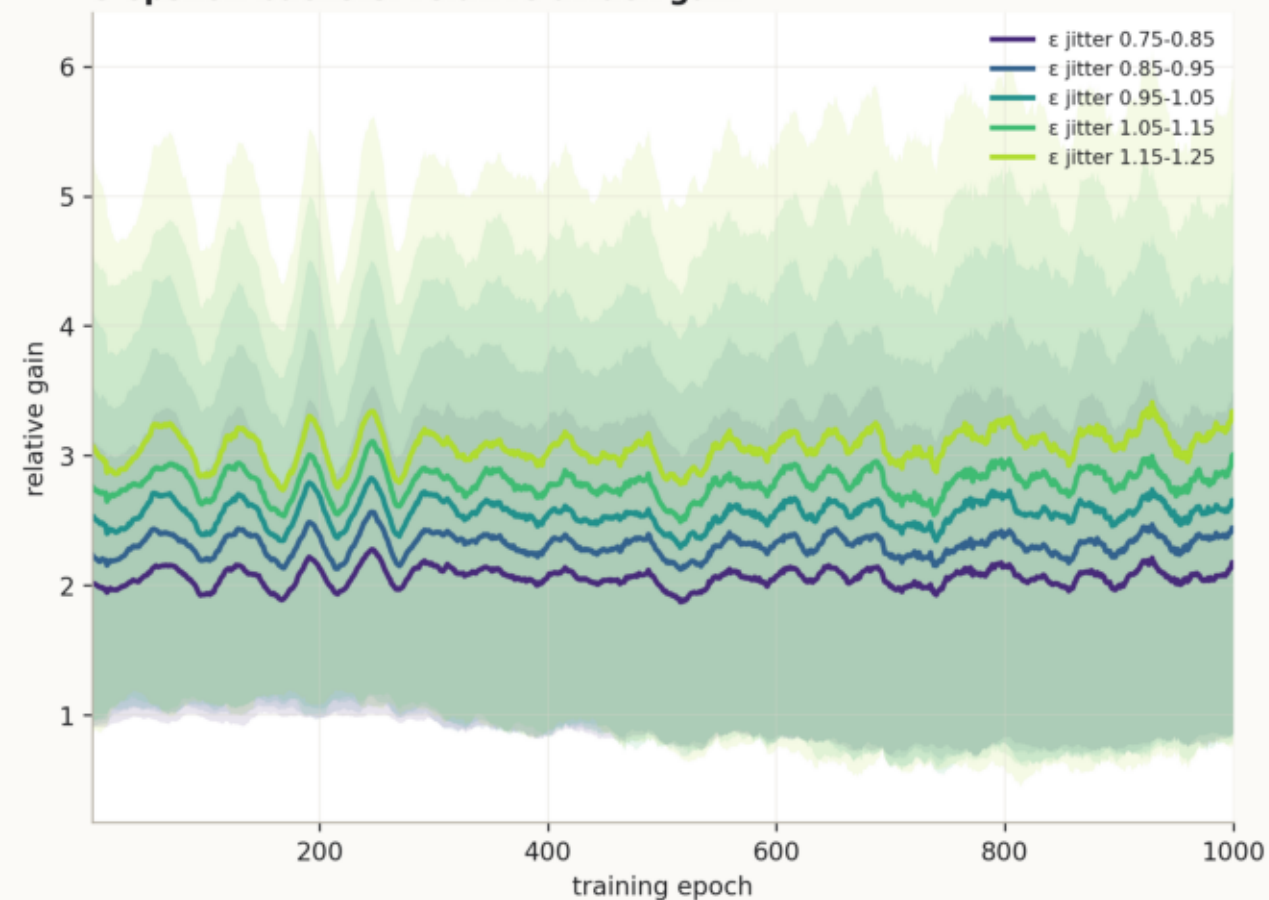


Attack loss bucket analysis

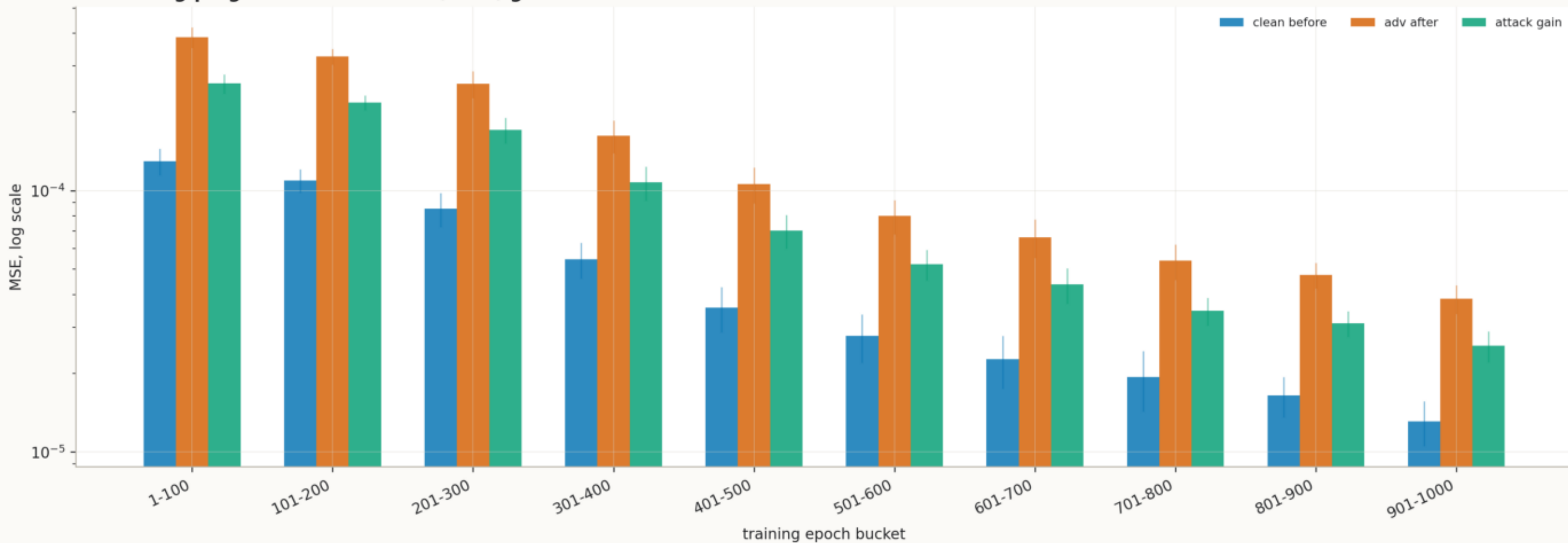
5 epsilon buckets: attack loss gain



5 epsilon buckets: relative attack gain



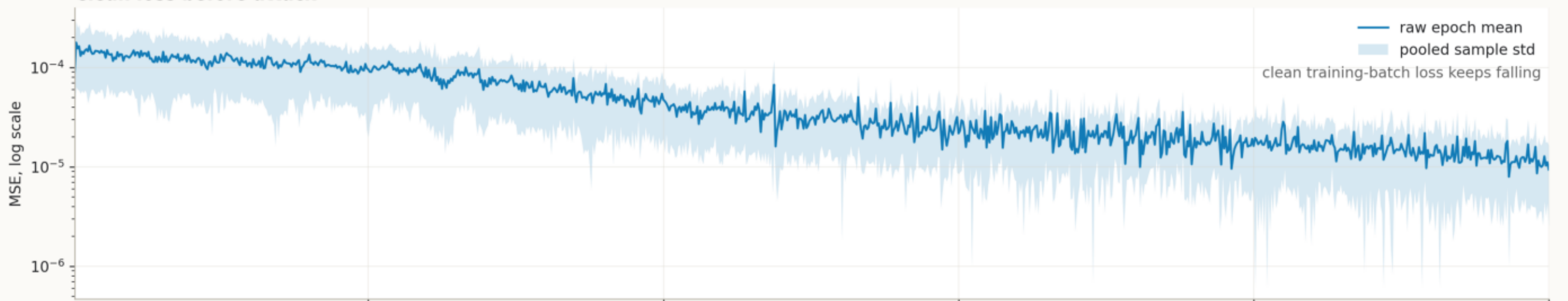
10 training-progress buckets: clean / adv / gain



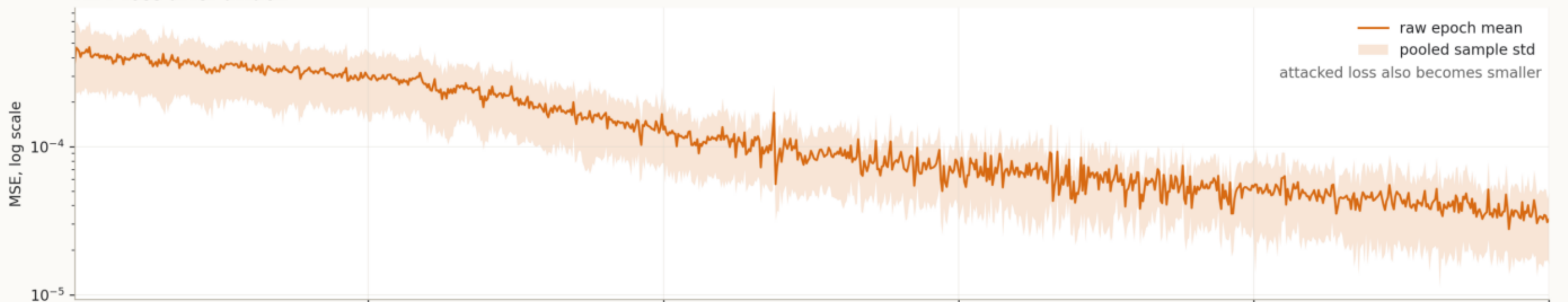
Raw attack loss progress, no moving average

Solid line is the raw per-epoch mean; shaded band is pooled mean $\pm$ std from epsilon buckets.

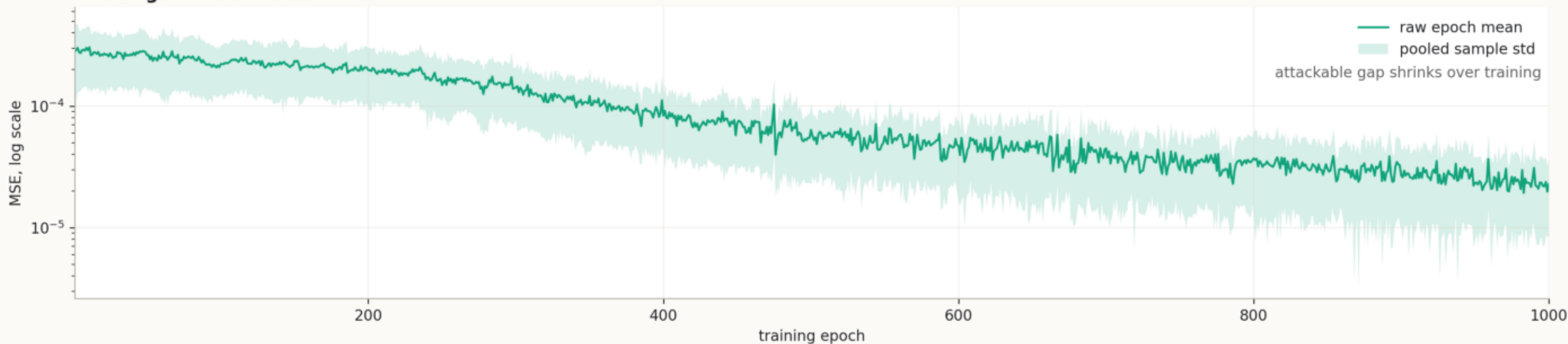
Clean loss before attack



ADV loss after attack



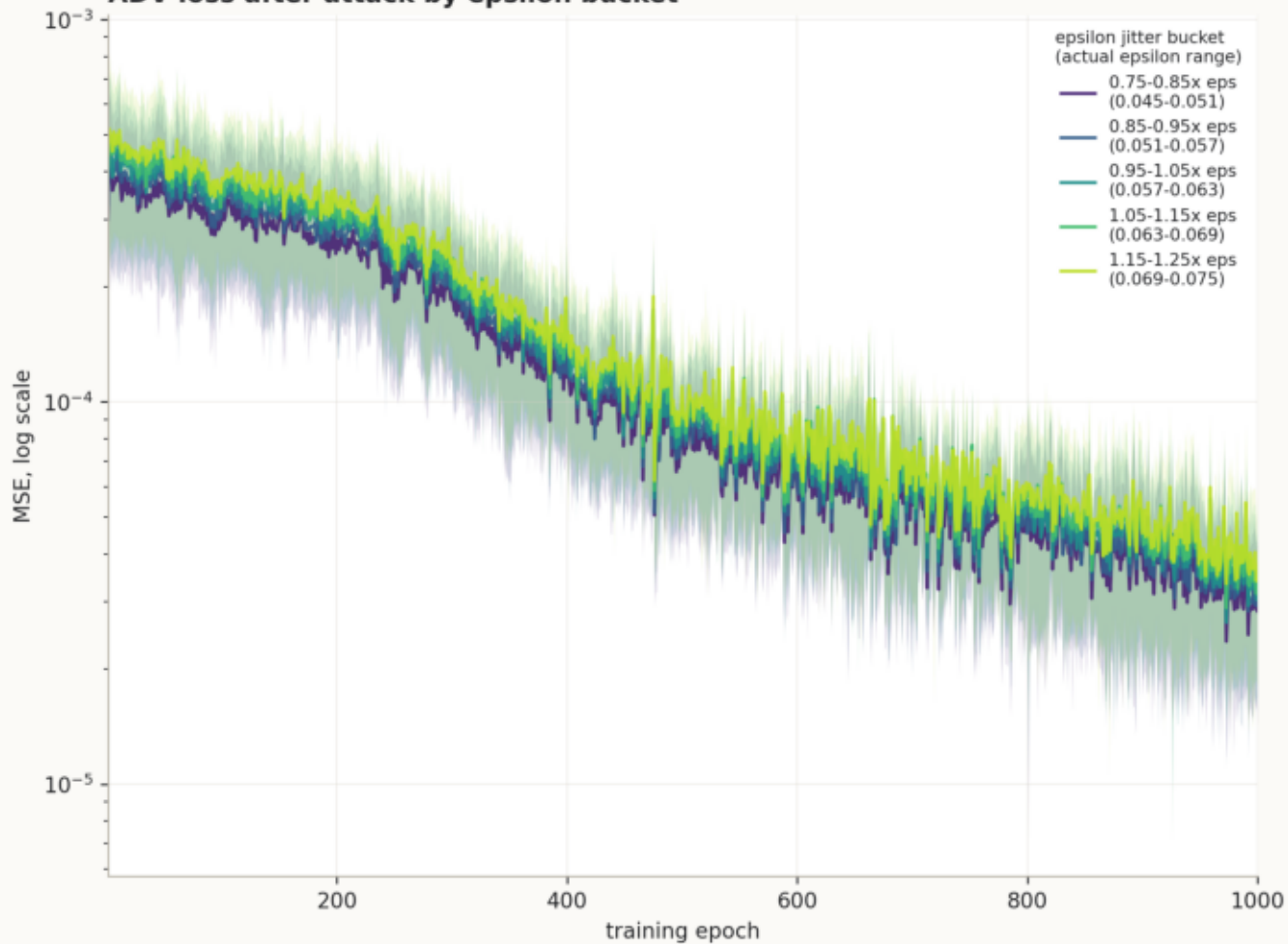
Attack gain = adv - clean



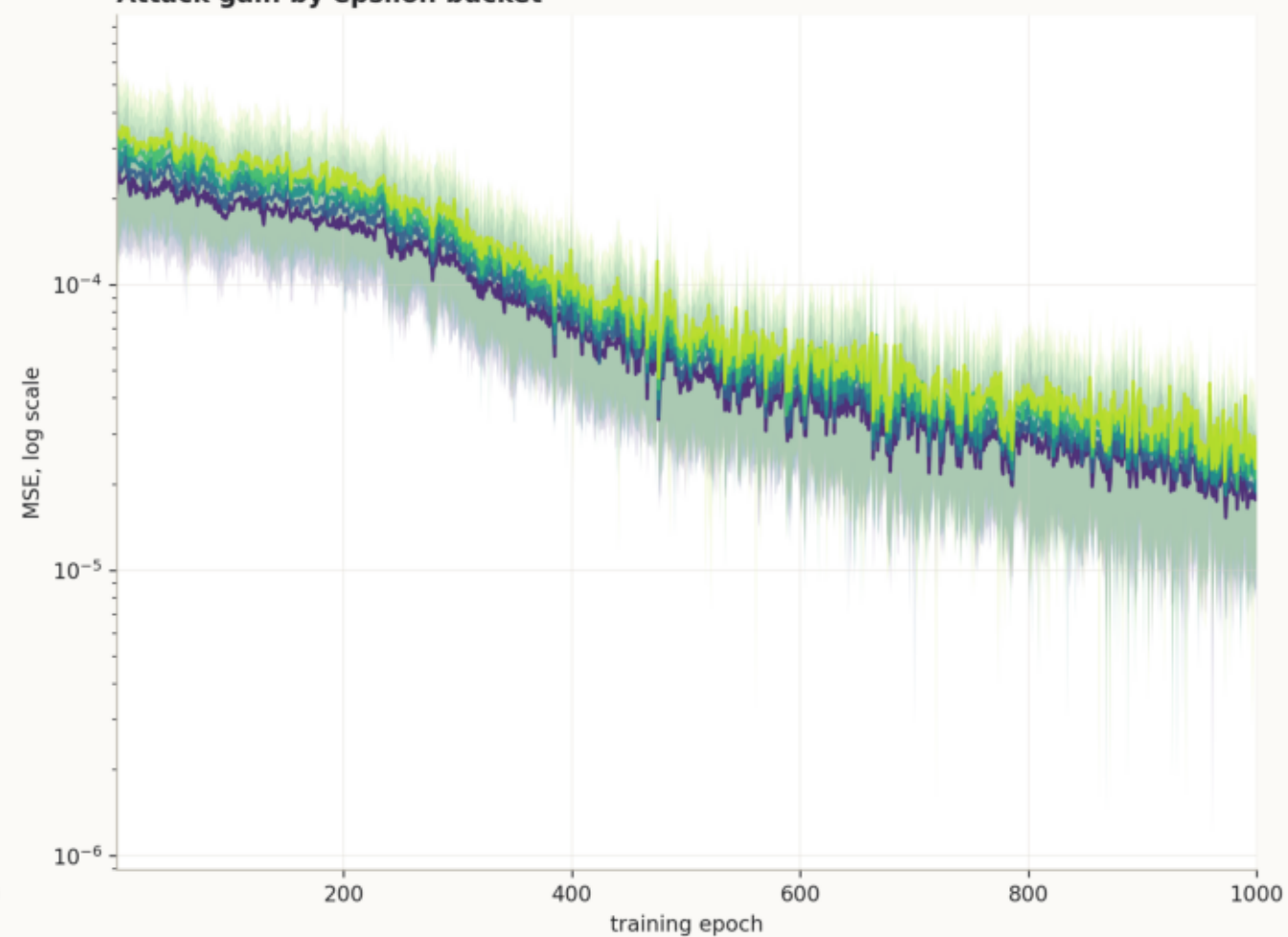
Raw epsilon-bucket attack curves with standard-deviation shading

No moving average: every line is the raw epoch mean for that bucket; shading is mean $\pm$ std inside the bucket.

ADV loss after attack by epsilon bucket



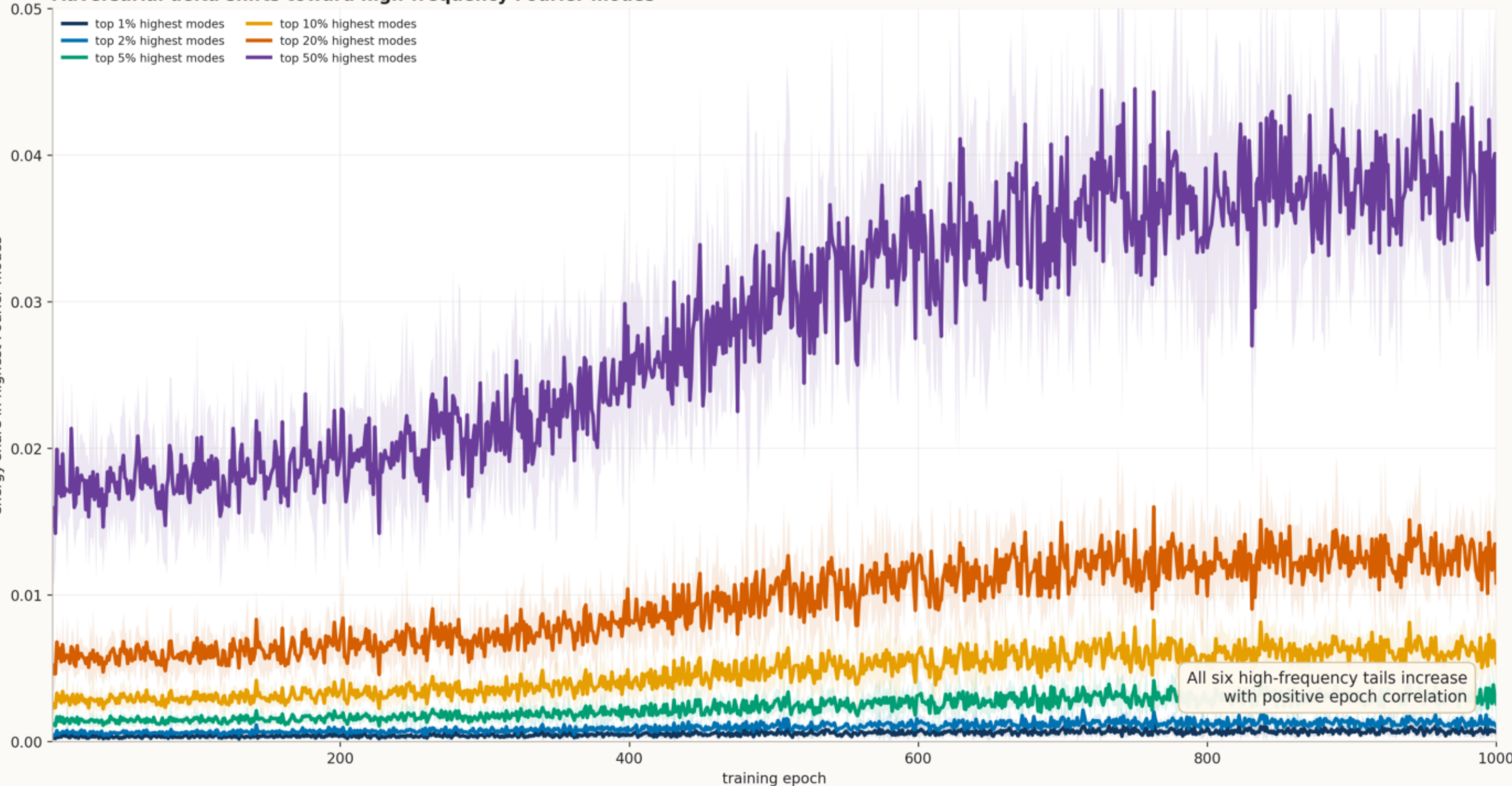
Attack gain by epsilon bucket



Adversarial delta shifts toward high-frequency Fourier modes

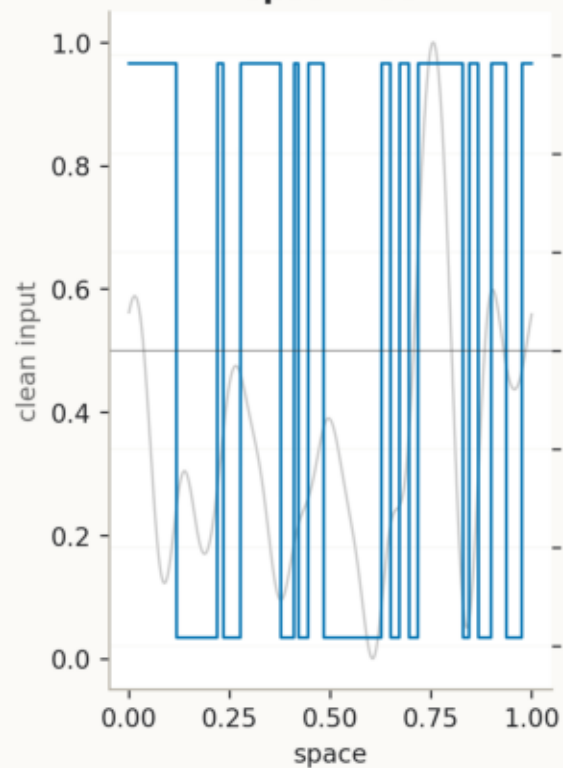
- top 1% highest modes
- top 2% highest modes
- top 5% highest modes
- top 10% highest modes
- top 20% highest modes
- top 50% highest modes

energy share in highest Fourier modes

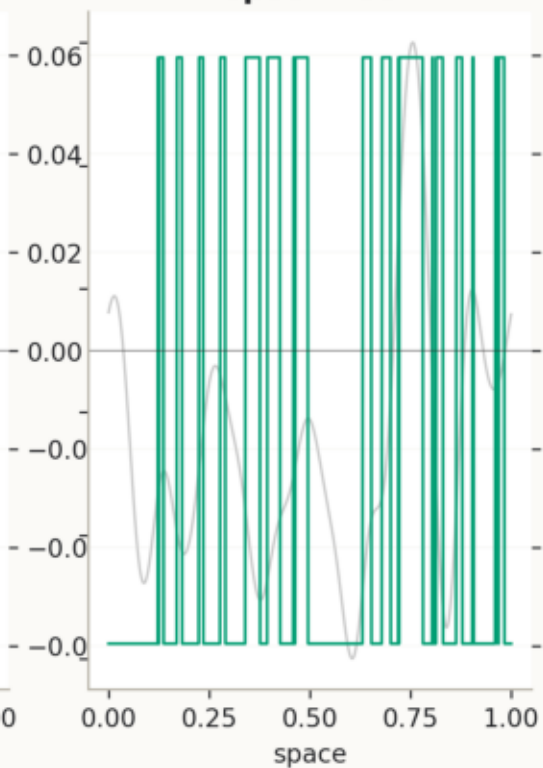


Fixed probe delta checkpoint view: probe 0, source index 0

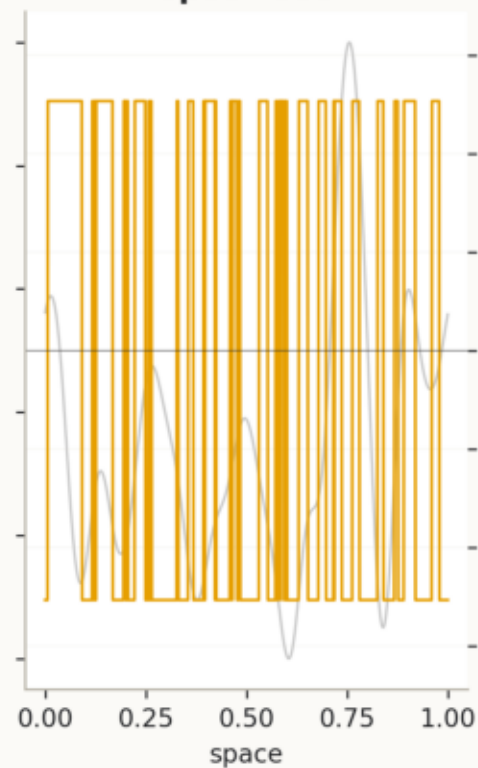
epoch 200



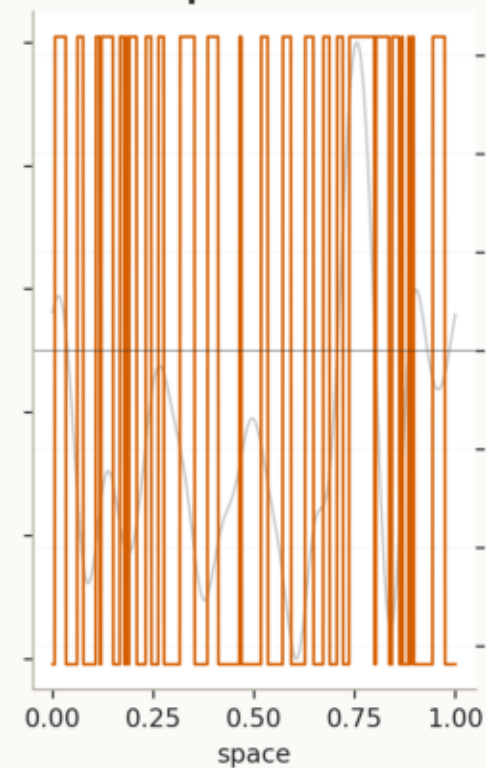
epoch 400



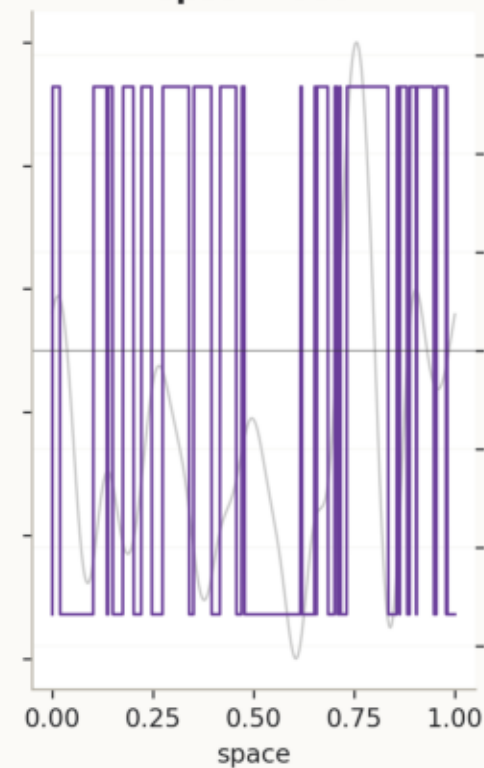
epoch 600



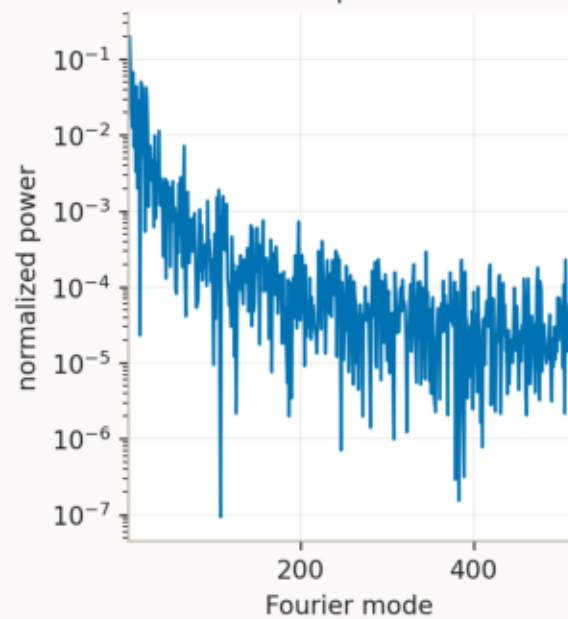
epoch 800



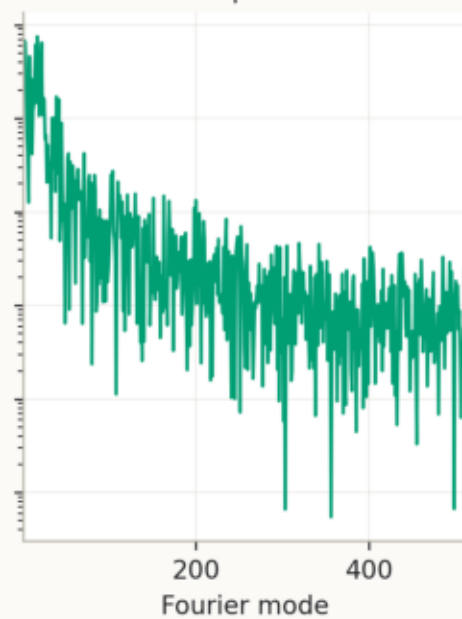
epoch 1000



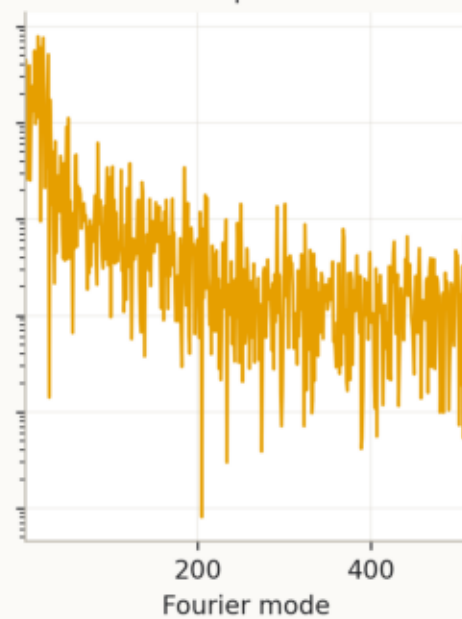
FFT power



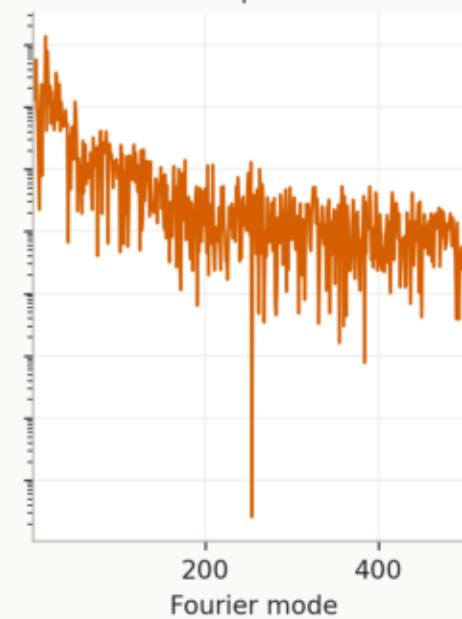
FFT power



FFT power



FFT power



FFT power

